2 **(a)**

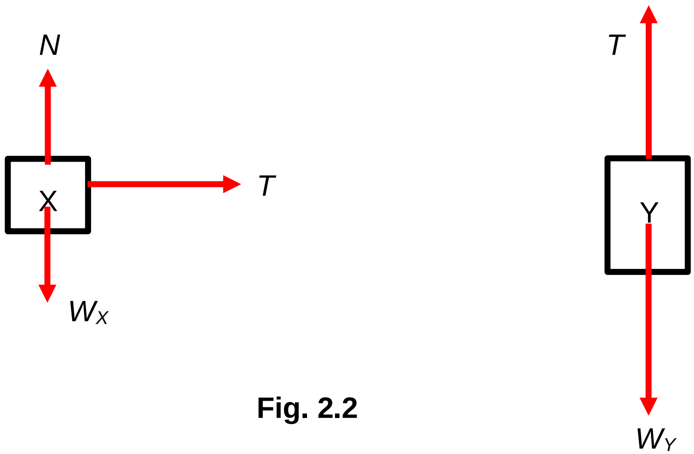

Fig. 2.2

B2

T: tension of string
N: normal contact force on X
W_X: weight of X
W_Y: weight of Y

1 mark for correct FBD for each mass

(b) **(i)** For X,
$T = m_X a$ --- (1)

For Y,
$W_Y - T = m_Y a$ --- (2) C1

Solving (1) and (2),
$0.600 \times 9.81 = (0.220 + 0.600)\ a$ A1
$a = 7.2 \text{ m s}^{-2}$

(ii) $T = 0.22 \times 7.2 = 1.58$ N B1

(iii) Resultant force = $0.600 \times 9.81 = 5.89$ N B1

(c) The collision is not perfectly inelastic at there are external forces acting on either object such as tension force on X and the force between the fixed block at table top. B2

Need to identify the correct external forces acting on the objects

2 (a) (i) resultant force on frame is zero and B1